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MILITARY SPECIFICATION

SURVIVAL KIT CONTAINERS, AIRCRAFT SEAT,
GENERAL SPECIFICATION, FOR

This specification is approved for use by all Departments and Agencies. of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for rigid seat style survival kit containers for aircraft equipped with ejection seats.

1.2 Classification. The survival kit containers shall be furnished in the following types as specified (see 3.5 and 6.2):

Type I - Operation of deployment sensing cable assembly caused by man separating from seat

Type II - Operation of deployment sensing cable assembly caused by canopy risers deploying from parachute pack.

APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal: form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-383	Plastic Material, Polyester Resin, Glass Fiber Base, Low Pressure Laminated

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-S-5002	Surfaces Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems
MIL-W-5625	Webbing, Textile, Nylon, Tubular
MIL-C-5688	Cable Assemblies: Aircraft, Proof Testing and Prestretching of
MIL-C-6021	Castings, Classification and Inspection of
MIL-T-6117	Terminal-Cable Assemblies, Swaged Type

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Aeronautical Systems Division, ASD/ENESS, Wright-Patterson AFB, Ohio 45433, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1660

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MIL-W-6858	Welding, Resistance: Aluminum, Magnesium, Nonhardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys; Spot and Seam
MIL-R-7575	Resin, Polyester, Low-Pressure Laminating
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aerospace Vehicles
MIL-T-8S04	Tubing, Steel, Corrosion-Resistant (304), Aerospace Vehicle Hydraulic Systems, Annealed, Seamless and Welded
MIL-W-8611	Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys; Process for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-R-9300	Resin Epoxy, Low-Pressure Laminating
MIL-P-9400	Plastic Laminate Materials and Sandwich Construction, Glass Fiber Base, Low Pressure Aircraft Structural, Process Specification Requirements
MIL-S-9479	Seat System, Upward Ejection, Aircraft, General Specification for
MIL-F-21840	Fasterner Tapes, Hook and Pile, Synthetic
MIL-S-27332	Seat Cushion Insert, Polyurethane Foam, Plastic, General Specification for
MIL-W-38283	Webbing, Textile, Aromatic Polyamide, Nonmelting
MIL-C-83124	Cartridge Actuated Devices/Propellant Actuated Devices, General Specification for
MIL-W-83420	Wire Rope, Flexible, for Aircraft Control
MIL-S-83862	Strap, Retention, Survival Kit
MIL-S-83863	Spacer Assembly, Survival Container Parachute Support
MIL-C-83864	Container, Inner, Survival Kit
MIL-C-83865	Cushion, Seat, Survival Container

STANDARDS

FEDERAL

FED-STD-406	Plastics: Methods of Testing
FED-STD-595	Colors

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of

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MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-846	Escape System Testing: Ground, Track, and Flight Test
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment: Requirements for
MIL-STD-889	Dissimilar Metals
MS17983	Compass - Magnetic, Pilots Standby

DRAWINGS

AIR FORCE

50B3502	Cable-Valve Release, Multiplace Life Raft
62E4336	Pattern-Life Raft Retainer, Adjustable
64C43333	Strap Assembly, Adjustment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following drawings form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue effect on date of invitation for bids or request for proposal shall apply.

USS Chemicals Div. (Code Identification 29807)

RP 755-32	Dropline Assembly

General American Research Division (Code Identification 19190)

1462C6000	Cartridge, 4 Sec Delay

(Copies of these drawings can be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

First article. This specification makes provisions for first article tests (see 4.4 and 6.2).

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3.2 Major components. Unless otherwise specified by the procuring activity (see 6.2), the survival kit containers shall include all of the major components listed on figure 1 and described herein, (Note: The restraint provision of the basic container, the retention straps and the deployment sensing cable assembly are listed but not shown on figure 1.)

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Nonmagnetic materials shall be used for all components except where magnetic materials offer definite advantages. Mateerials that are not susceptible to fungus attack shall be used to the greatest extent practicable. Materials that are susceptible to fungus attack, if used, shall be treated with a fungicidal agent. The materials used in the construction of the components of the survival kit containers shall be the lightest possible weight consistent with configuration, strength, and any other requirements specified herein.

3.4.1 Metals. Metals shall be corrosion resistant or suitably treated to resist corrosion due to fuels, salt spray, or atomospheric conditions likely to be met in storage or normal service.

3.4.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

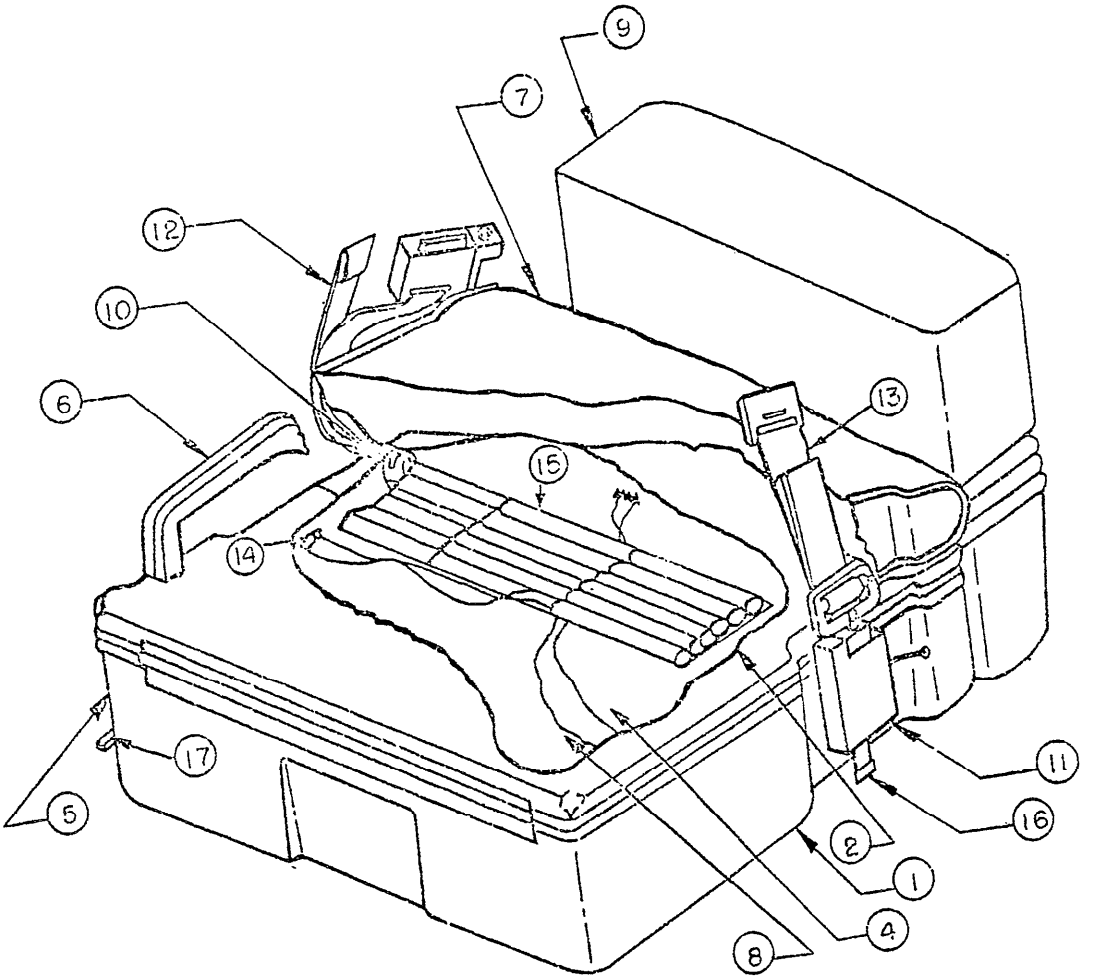
3.4.2 Nometallic materials. The use of nonmetallic materials shall require approval the procuring activity, for the specific application involved.

3.4.2.2.1 Fiber glass material. If fiber glass material is used in the construction of the basic container, it shall be manufactured in accordance with the requirements of MIL-P-9400. The fiber glass material shall meet the physical requirements of table I of this specification and the workmanship requirements of L-P-383, Glass cloth, if used, shall be preimpregnated cloth. Resins shall be in accordance, with MIL-R-7575 or MIL-R-9300 for polyester resin or epoxy resin, respectively.

3.4.3 Protective treatment. If materials that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage are used in the construction of the basic container, the materials shall be protected against such deterioration in a manner that will not in any way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extreme climatic and environmental conditions shall be avoided.

3.4.3.1 Aluminum parts. All aluminum parts shall be anodized in accordance with MIL-A-8625. All exposed parts shall be processed after anodizing with a potassium bichromate seal and dyed a neutral color suitable to the procuring activity.

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1 COMPARTMENT (LOWER CONTAINER OF BASIC CONTAINER)
2 LID (UPPER CONTAINER OF BASIC CONTAINER)
3 RESTRAINT PROVISIONS OF BASIC CONTAINER (NOT SHOWN)
4 INNER CONTAINER
5 AUTOMATIC ACTUATOR (INCLUDING CARTRIDGE)
6 OPERATING HANDLE (EMERGENCY RELEASE)
7 CUSHION
8 RAFT RETAINER
9 PARACHUTE SUPPORT
10 RIGHT-HAND HARNESS ATTACHMENT STRAP SIDE RELEASE AND LID LOCK ASSEMBLY
11 LEFT-HAND HARNESS ATTACHMENT STRAP SIDE RELEASE AND LID LOCK ASSEMBLY
12 RIGHT-HAND ATTACHMENT STRAP
13 LEFT-HAND ATTACHMENT STRAP
14 DROPLINE ASSEMBLY
15 DROPLINE RETAINERS
16 DROPLINE CONNECTING PLUNGER DEVICE
17 OVERRIDE SELECTOR DEVICE
18 RETENTION STRAPS (NOT SHOWN)
19 DEPLOYMENT SENSING CABLE ASSEMBLY (NOT SHOWN)

FIGURE 1. Survival kit container.

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TABLE I. Physical properties of fiber glass

Characteristics	Cloth	Mat and Roving
Flex strength, pounds per square inch (psi), minimum	60,000	30,000
Flex modulus, minimum	3.0×10^6	1.1×10^6
Tensile strength, psi, minimum	47,000	22,000
Comprehensive strength, psi, minimum	45,000	20,000
Moisture absorption, percent per 24 hours, maximum	0.5	1.0
Impact strength, unnotched, foot-pounds per inch, minimum	25	20

3.4.3.2 Steel parts. All steel parts shall be cadmium or zinc plated in accordance with the applicable specifications unless the part is made of corrosion-resistant steel or has-been othewise protected in accordance with MIL-S-5002.

3.5 Design. The type I survival kit container shall be designed to be attached to the restraint subsystem and to stay with the user during ejection and parachute descent. The type II survival kit container shall be designed to be attached to the personnel torso harness of the parachute and to stay with the user during ejection and parachute descent. Unless otherwise specified herein, the type I and the type II survival kit containers shall be designed for aircraft ejection seats that conform to MIL-S-9479. (see 6.2). The main difference between the type I and the type II survival kit containers is the operation of the deployment sensing cable assembly, which is based upon the operation of the parachute recovery subsystem of the ejection seat, and is as follows: The deployment sensing cable assembly of the type I survival kit container shall operate the automatic actuator by a force in the range of 2 to 10 pounds as the survival kit container separates approximately 30 inches from the seat (seat-man separation accomplished by inflation of the recovery parachute in accordance with MIL-S-9479). The deployment sensing cable assembly of the type I survival kit container shall be automatically disconnected after operation of the seat emergency restraint subsystem release. The deployment sensing cable assembly of the type II survival kit container shall be operated automatically by a force not to exceed 50 pounds, measured at the canopy risers, as the canopy risers deploy from the parachute pack (seat-man separation not accomplished by inflation of the recovery parachute in accordance with MIL-S-9479).

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3.5.1 Operation The following operations shall be allowed by the survival kit containers.

(a) Removal of the survival kit container from the aircraft for maintenance: The manual override selector lever of the automatic actuator shall be used as the safety mechanism of the actuator firing pin during maintenance handling to prevent accidental automatic operation. . Disconnection of all attachments of the survival kit container to the seat, to the parachute, and to the aircraft shall be made without the use of special tools and without opening the survival kit container.

(b) Normal egress from the aircraft after a flight, leaving the survival kit container in the aircraft: Disconnection of the survival kit container shall be made at the personnel torso harness accessory rings, The automatic. deployment sensing cable assembly shall not require separate disconnection for normal egress.

(c) Emergency egress from the aircraft on the ground or in the water: Operation of the personnel restraint subsystem will release the man from the seat. The survival kit container may be removed or may remain in the aircraft based upon the user's decision. When the survival kit container is to be left in the aircraft, the emergency release handle shall permit manual actuation of" the survival kit container to release all connections between the survival kit container, the user, and the parachute. The raft shall not be actuated. The survival kit container dropline attachment fitting shall remain with the portion of hardware remaining with the survival kit container. The automatic actuator deployment sensing cable shall separate automatically as required based upon the ejection seat system emergency egress mode.

(d) Ejection with subsequent manual or automatic release of the survival kit container from the man followed by inflation of the raft. The raft and the other survival equipment shall be secured to the man by means of a dropline. This operation shall be accomplished in the following steps:

(1) Separation of the survival kit container from the seat shall automatically connect the dropline to the portion of the survival kit container hardware remaining attached to the personnel torso harness accessory ring after survival kit container deployment. The parachute canopy will automatically deploy and fully inflate. As the man separates from the seat (type T survival kit container) or as the canopy risers deploy from the parachute pack (type II survival kit container), the deployment sensing cable assembly shall be pulled and shall separate and initiate the automatic actuator and cause survival kit container hardware operation 4 seconds (plus or minus 1 second) after sensing cable operation unless the override selector prevents operation.

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(2) Automatic actuation or a manual pull of the survival kit container operating handle shall separate the survival kit container from the man and unlock the lid. The upper end of the dropline shall be retained with the personnel torso harness. The LRU-16/P raft inflation system shall be actuated by dropline deployment. The inner container and the lower container shall be retained at the bottom of the dropline.

3.5.2 Survival equipment. The survival kit containers shall be designed to accommodate the survival equipment specified herein. Unless otherwise specified by the procuring activity (see 6.2). the survival equipment for each survival kit container shall consist of an LRU-16/P raft, a sleeping bag, and an inner container that contains other survival equipment. Except for the inner container, this survival equipment shall not be furnished with the survival kit container; however, provisions for packing this survival equipment in the basic container and for inflating the raft shall be made,

3.6 Construction. The basic container of each survival kit container shall be constructed so that parts will not work loose in service and so that it will withstand the strains, jars, and any other conditions incident to shipment, storage, installation, and service without failure. All hardware of the survival kit container that could come in contact with the inner container and the raft during deployment and during service operation shall be smooth and free of sharp edges or protrusions.

3.6.1 Method of assembly. Welding riveting, or any other permanent methods of assembly may be used provided that disassembly or reassembly of the component for inspection or overhaul purposes can be accomplished rapidly, safely, and without damage to the assembly. Operation after assembly shall not be dependent upon selective fit of any or all conponents, Welding shall be accomplished in accordance with MIL-W-8611 or MIL-W-6858 or both. Soldering and btrazing shall be done accordance with current applicable specifications. Welding, soldering, and brazing shall be carefully performed to ensure that all joints have proper penetration, proper bead, and proper strength. Riveting operations shall be carefully performed to ensure that the rivets are tight and satisfactorily headed.

3.6.2 Castings. Castings shall be designed and inspected in accordance with MIL-C-6021. Castings shall not be used in applications where failure could result in loss of the survival kit container during ejection,

3.6.3 Safetying. All threaded parts shall be positively locked by self-locking nuts or any other approved methods. Safety wire shall not be considered an approved method.

3.6.4 Swaging Swaging shall be done in accordance with MIL-T-6117 and shall be carefully performed. Each swaging operation shall be designed to withstand 100 pounds or three times the maximum service load, whichever is greater.

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3.6.5 Threads. Except for permanently installed pipe threaded plugs, or as necessitated by functional purposes and unless otherwise specified in the applicable documents, only straight threads conforming to Unified Thread Series, class 3A or 3B of MIL-S-7742 shall be used.

3.6.6 Control cables. The control cables shall be fabricated of cable that conforms to MIL-W-83420 and shall be tested and prestretched in accordance with MIL-C-5688. Any control cable adjustments necessary for manufacturing shall be permanently locked after rigging. The control cables shall be properly installed. Where necessary to prevent inadvertent actuation or damage to the control cables by the contents or flexure of the survival kit container, the control cables shall be enclosed in tubing that conforms to MIL-T-8504. All control cables shall be properly guarded to prevent wearing or fraying during the continued use of the survival kit container. Each cable shall be installed in such a manner that the cable is never bent in a radius that is smaller than five times the nominal diameter of the cable.

3.7 Performance

3.7.1 Environmental conditions. The hardware of the survival kit container shall operate satisfactorily as specified in 3.7.2.1 and 3.7.2.2 under the following environmental conditions:

(a) While the survival kit container is subjected to a temperature of plus 160 degrees Fahrenheit

(b) After the survival kit container has been subjected to temperature shock

(c) While the survival kit container is subjected to a temperature of minus 65 degrees Fahrenheit

(d) During and after exposure of the survival kit container to dust

(e) During and after exposure of the survival kit container to humidity

[f) During and after exposure of the survival kit container to salt atmosphere

(g) During and after exposure of the survival kit container to vibration

(h) After the survival kit container has been subjected to shock.

3.7.2 Hardware operation

3.7.2.1 Parachute descent. When the operating handle is activated by the force specified in 3.8.4.1 while in the parachute descent mode, the hardware

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of the survival kit container shall accomplish the following actions:

(a) Separate the survival kit container from the man

(b) Open the lid locks

(c) Actuate the raft by dropline deployment

(d) Couple the dropline to the parachute harness attaching hardware.

3.7.2.2 Emergency ground egress. When the force specified in 3.8.4.1 is applied to the operating handle during emergency ground egress, the hardware of the survival kit container shall accomplish the following actions:

(a) Separate the survival kit container from the man

(b) Open the lid locks

(c) Separate the dropline as required.

3.7.2.3 Automatic operation. When the deplopment sensing cable assembly is activated by the applicable force (2 to 10 pounds for the type I survival kit container or a maximum of SO pounds at the canopy risers for the type II survival kit container) applied in any direction, the distal cable end shall separate and the automatic actuator shall be operated. The deployment sensing cable assembly shall release the actuator firing pin to fire the cartridge, The hardware of the survival kit container shall operate 4 seconds (plus or minus 1 second) after the automatic actuator has fired the cartridge. It shall not be possible to install the right side attachment hardware if the actuator has not been reset after an operation. The deployment sensing cable assembly shall resist the effects of service conditions so that accidental actuation or failure due to wear or interference from associated objects shall not occur.

3.7.3 Interchangeability of critical parts. All critical parts of the survival kit containers such as the lids, the cushions of the same part number, the operating handles, and the survival kit container attaching hardware shall be interchangeable in accordance with MIL-I-8500.

3.7.4 Proof loads. After the proof loads specified in 3,7.12.1.1, 3,7.12.2 and 3.7.12.4 have been applied to the survival kit container, the survival kit container shall operate as specified in 3.7.2.1 and 3.7.2.2 and shall not be damaged nor permanently deformed.

3.7.5 Lid retention. The lid locking latches shall unlock automatically when either the automatic actuator or the operating handle is activated. The lid retaining mechanism shall not permit the lid to separate from the compartment

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when an internal load of 700 pounds is applied to the lid. There shall be no visible evidence of failure of the lid locks. After an iternal load of 700 pounds has been applied to the lid, the hardware of the survival kit container shall operate satisfactorily as specified in 3,7.2.1 and 3.7.2,2,

3.7.6 Fiber glass container. If the basic container is made of fiber glass material, the fiber glass material of the basic container shall conform to the requirements specified in 3.4.2.1.

3.7.7 Parachute opening shock. After the survival kit container has been subjected to a simulated parachute opening shock of at least 30 G (peak) and a minimum rate of increase to 500 G per second, the hardware of the survival kit container shall operate as specified in 3.7.2.1 and 3.7.2.2.

3.7.8 Pachute drop. The following functions shall occur when the survival kit container is subjected to parachute drops, simulating operational use, at speeds and altitudes varying from zero knot at 2,000 feet altitude through 400 knots at 18,000 feet altitude (parachute deployment 1 second after aircraft exit) .

(a) The actuator shall operate the hardware of the survival kit container.

(b) The type I survival kit shall deploy on its dropline 4 seconds (plus or minus 1 second) after seat-man separation. The type II survival kit container shall deploy on its dropline 4 seconds (plus or minus 1 second) after the canopy risers deploy from the parachute pack.

3.7.9 Windblast When the survival kit container is subjected to windblast of 600 knots at sea level or equivalent, the lid shall not separate from the survival kit container, and the operating handle shall remain on the survival kit container After the survival kit container has been subjected to windblast of 600 knots at sea level or equivalent, the hardware of the survival kit container shall operate satisfactorily as specified in 3.7.2.1 and 3.7.2.2.

3.7.10 Magnetic interference. The survival kit container shall not deflect a pilot's standby compass (MS17983 or equivalent) more than 3 degrees when the survival kit container is placed 6 inches from the face of the compass.

3.7.11 Adjuster

3.7.11.1 Adjuster creep. The webbing shall not move through the adjuster more than 1/2 inch when alternating loads of 20 pounds and O pound are applied for 30 cycles in each of the following five directions: normal, 60 degrees forward 60 degrees aft. and 60 degrees to either side of normal to the survival kit container lid.

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3.7.11.2 Release of adjuster. With an initial load of 20 pounds of tension applied to the attachment webbing, the adjuster shall allow complete slippage of the webbing when “a 30-pound release load is applied to the slider assembly.

3.7.12 Static load

3.7.12.1 Parachute attaching hardware

3.7.12.1.1 Hardware and container section. The parachute attaching hardware and the survival kit container section to which it is attached shall withstand load conditions of 1,400 pounds (proof) without permanent deformation and 2,100 pounds (ultimate) for each side.

3.7.12.1.2 Handle operation. With a load of 100 pounds applied to the parachute attaching hardware and to each side of the survival kit container hardware, hardwarea pull of less than 40 pounds applied 2 inches from the forward end of the handle and moved through the arc of the handle travel shall result in operation of the handle and release of the survival kit container hardware under emergency ground egress and parachute descent operating conditions.

3.7.12.2 Container (seating section). The seating section of the basic container shall not show any permanent deformation after the survival kit container has been subjected to a compression load of 4,500 pounds proof. The ultimate strength of the seating section of the basic container shall be 5,600 pounds.

3.7.12.4 Handle side load. When the operating handle is installed in the normal locked position and an outward proof load of 500 pounds is applied to the operating handle, the operating handle shall not be permanently deformed.

3.7.12.4 Handle lock load. When a proof load of 200-pound tension is applied to the grip of the operating handle, the operating handle and the lock shall not be permanently deformed.

3.7.12.5 Dropline connecting plunger device. The dropline connecting plunger device shall be fully retracted by the weight of the survival kit container when the survival kit container is placed flat against the bottom of the ejection seat pan and weighted at the center of the survival kit container to achieve a total weight of 30 pounds. When the dropline connecting plunger device is extended, it shall withstand a force of 200 pounds in any horizontal direction without damage. When the dropline connecting plunger device is free, it shall extend 1-1/2 inches (plus or minus 1/8 inch) from the bottom of the survival kit container. The changing of modes shall occur while the dropline connecting plunger device is extended in the range of 1-1/8 inches (plus or minus 1/8 inch) from the bottom of the survival kit container.

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3.7.13 Reliability. The survival kit container includes a redundancy feature of an automatic operating mode and of a manual operating mode. The reliability of the automatic actuator shall be in accordance with MIL-C-83124.

3.7.13.1 Manual operating mode. The operating hardware of the survival kit container in the manual operating mode shall have a mean cycles between failures (MCBF) of 100 cycles with a minimum reliability of 0.99 at a 90 percent confidence level. The following functions shall occur during the manual operating mode:

(a) The dropline connecting plunger device shall be released from its depressed position; it shall extend and connect the dropline to the harness attachment strap assembly,

(b) The sensing cable of the type I survival kit container shall fully extend and then be operated and separated from the automatic actuator by a force in the range of 2 to 10 pounds, applied in any random direction. The sensing cable of the type II survival kit container shall fully extend and then be operated and separated from the automatic actuator by a force not to exceed 50 pounds, measured at the canopy risers, applied in any random direction.

(c) The automatic actuator shall not be operated.

(d) The handle shall be unlocked by a force of 20 pounds (plus or minus 5 pounds), operated by a force not to exceed 30 pounds (applied 2 inches from the front), and then shall separate.

(e) Simultaneously with handle separation, both lid locks shall disengage, and both harness attachment strap side releases shall separate.

3.7.13.1.1 Longevity. The survival kit container shall have a service operating life span of not less than 300 cycles in the manual operating mode before wearout failures occur or the equipment consistently fails to meet the specified MCBF index.

3.7.13.2 Automatic operating mode. The automatic operating mode of the hardware of the survival kit container shall have a demonstrated reliability of 0.97 at a 90 percent confidence level. The following functions shall occur during the automatic mode:

(a) The dropline connecting plunger device shall be released from its depressed position and shall extend and connect the dropline to the harness attachment strap assembly,

(b) The sensing cable of the type I survival kit container shall fully extend and then be operated and separated from the automatic actuator

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by force in the range of 2 to 10 pounds, applied in any random direction simulating seat-man separation. The sensing cable of the type 11 survival kit container shall fully extend and then be operated and separated from the automatic actuator by a force not to exceed 50 pounds at the canopy risers, applied in any random direction, simulating deployment of the canopy risers from the parachute pack.

(c) The automatic actuator shall fire the cartridge.

(d) Both lid locks shall disengage and both harness attachment strap side releases shall separate within 4 seconds (plus or minus 1 second) after sensing cable operation.

3.7.14 Maintainability. The manufacturer of the survival kit containers shall establish a maintainability program and submit reports thereon in accordance with MIL-STD-470. Unless otherwise specified by the procuring activity (see 6.2), the survival kit containers shall have a median corrective maintenance downtime of 30 minutes.

3.8 Details of components

3.8.1 Basic container. The basic container shall consist of a lower container (compartment) an upper container (lid), and restraint provisions that will hold the packed survival kit container in the ejection seat during negative G maneuvers. The basic container shall provide a maximum interior volume consistent with the requirements specified herein. The bottom of the basic container shall have an absorbent non-skid surface that is 1/16 inch thick. Where it is necessary to provide holes in the basic container, there shall be no interstices greater than 1/16 inch. This requirement shall also apply to the lid parting line with the basic container packed with equipment. 411 mating edges of the compartment and the lid shall have aluminum extrusion-s.

3.8.1.1 Compartment. The compartment shall provide a minimum of 1,800 cubic inches of usable space that will accommodate survival equipment that will weigh a maximum of 40 pounds (see 3.5.2).

3.8.1.2 Lid. The lid shall be flat or contoured as specified by the procuring activity (see 6.2); however. only contour patterns that have been previously approved by the procuring activity may be used. The basic lid plane that forms the seat surface shall be raised 4 to 6 degrees from the aircraft horizontal line toward the front of the survival kit container. The lid shall have cushion attachment fasteners in a pattern that is compatible with the cushion specified in 3.8.5 and that is capable of retaining the cushion to the lid during the ejection sequence. The forward edge of the lid shall have a separating, hinge. The lid shall have two locking latches, one on each side, which shall be incorporated into the side release hardware. The locking latches shall incorporate indicators to show lock engagement. It shall not be possible to accidentally open the lid during normal inflight usage or during bailout. Unless otherwise specified by the procuring activity (see 6.2), the lid shall be retained to the dropline during survival kit container deployment.

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3.8.1.3 Restraint provisions. The survival kit shall be retained in the ejection seat bucket within 0.75 inch of the bottom during negative G aircraft flight maneuvers and shall separate freely during the seat-man separation phases of ejection seat operation.

3.8.2 Inner container. The inner container shall conform to MIL-C-83864 except that the size and the shape of the inner container may vary to fit the compartment of the basic container.

3.8.3 Automatic actuator. The automatic actuator shall be designed as a cartridge actuated device in accordance with MIL-C-83124, shall be constructed so that parts will not work loose in service, and shall meet the requirements specified herein. The automatic actuator shall have a cartridge to 'control actuation with respect to time after parachute deployment and seat separation and to provide the power to operate the hardware of the survival kit container. The automatic actuator shall be operated by a deployment sensing cable assembly that will release the actuator firing pin to fire the cartridge. When the automatic actuator is installed, it shall have no exposed moving parts.

3.8.3.1 Cartridge. The cartridge shall be a 4-second (plus or minus 1-second) time delay, percussion-type, expansible, closed system cartridge (a cartridge that deforms but does not rupture upon firing). The cartridge shall be equivalent to General American Research Div Drawing 1462C6000. The cartridge shall be installed to meet the following requirements:

(a) The cartridge firing pin shall be locked in its uncocked or unloaded position so that accidental firings shall not be possible.

(b) There shall be a visual indication that the cartridge is installed.

(c) The exterior of the survival kit container shall have a visual indication to show that the firing pin has or has not been reset from a previous operation.

[d) The cartridge shall be installed in a housing that is designed and constructed so that the cartridge cannot be positioned in any way that could prevent the proper operation of the automatic actuator.

3.8.4 Operating handle. The operating handle shall be a separating lever that incorporates a flexible closed loop grip and a suitable linkage-type locking device. A hook-cam lever. linkage-type locking device will be acceptable; a locking-ball-type mechanism shall not be acceptable. The color of the operating handle shall be yellow that conforms to 33538 of FED-STD-595. The following words shall be embossed in black letters on the top of the operating handle: EMERGENCY RELEASE. The operating handle shall be in front

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of the hardware, on the right side of the survival kit container. The operating handle shall be positioned so that it will not extend past the front of the survival kit container, so that it will not cause uncomfortable pressure points, and so-that the front of the operating handle will be not more than 2 inches from the front of the survival kit container, The operating handle shall be in an easily accessible position and easily operable both during flight and while the man is suspended in the parachute harness. Both ends of the operating handle shall be curved and faired into the survival kit container side or top to preclude the possibility of snagging or catching parachute lines, clothing, or straps on or under the operating handle. Provisions for lock wiring the operating handle to the lower container shall be made,

3.8.4.1 Operating force. When a pull of 20 pounds (plus or minus 5 pounds) is applied to the operating handle, an upward and aft rotation of not more than 4 inches (measured along the arc produced by the front of the operating handle) shall cause the lid of the survival kit container to unlock and shall cause the separation of the necessary components of the hardware of the survival kit container. When the operating handle is pulled while the survival kit container is in the seat, the dropline shall be connected to the portion of the hardware remaining with the survival kit container. Unless otherwise specified herein a maximum pull of 15 pounds shall be required to free the operating handle from the survival kit container under both parachute descent and emergency ground egress operating conditions. During and after the subjection of the survival kit container to the environmental conditions described under 3.7.1, a maximum pull of 30 pounds shall be required to free the operating handle from the survival kit container under both parachute descent and emergency ground egress operating conditions.

3.8.5 Cushion. Unless otherwise specified herein, the cushion shall conform to MIL-C-83865. When approved by the procuring activity, cushion insert materials that are superior in comfort, fire resistance, aging, endurance, attenuation, and thermal stability to the cushion insert materials specified in MIL-S-27332 may be used. The cushion shall be configured as required to provide sufficient support with clearances for operation and access by the aircrewman to the following: aircraft control column, ejection seat control handle, rudder pedals, personal services delivery lines, and survival kit contrainer operating hardware. The cushion shall be contoured to position the aircrewman at the proper seat reference point for optimal spinal alignment and back support during ejection as specified in MIL-S-9479 and shall not amplify the acceleration forces imposed upon the aircrewman beyond the limits specified therein when incorporated in the intended ejection system,

3.8.6 Raft retainer. The raft retainer shall conform to Drawing 62E4336.

3.8.7 Parachute support. The parachute support shall be a spacer assembly that conforms to MIL-S-83863.

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3.8.8 Harness attachment straps and hardware, The harness attachent straps and the hardware of the survival kit containers shall conform to figure 2.

3.8.8.1 Side release and lid lock assemblies. The harness attachment strap side release and lid lock assemblies shall be designed so that, when the operating handle is pulled, the locks will remain in the open position and the mating units at the hardware cannot be relocked until the operating handle is re-engaged on the survival kit container or until a secondary operation is performed. A visual indicatcr for inspection of a secure interlock engagement of the side hardware shall be provided,

3.8.8.2 Adjusters. The adjusters shall be located at the upper section of the separating container attachment fitting (see figure 2), The top of the adjuster shall not extend more than 1 inch above the top surface of the lid. Each adjuster shall be a quick-adjustable, spring-loaded, friction-type adjuster that will accept two thicknesses of the webbing specified in 3,8,8,3, The slider part of the adjuster shall have grips for easier loosening of the webbing while the survival kit container is in the ejection seat. The adjuster shall be operable with reasonable ease to take up slack in the linkage to provide safe retention of the survival kit container during bailout or ejection procedures.

3.8.8.3 Harness - attachment straps. The harness attachment straps shall be adjustable strap assemblies that conform to Drawing 64C43333 except that the webbing shall be sage green webbing (see 6.3) conforming to type IV of MIL-W-38283 and that the webbing shall be stitched along the perimeter only in the adjustable portion, The webbing end stop may be any suitable design that will prevent pulling of the webbing end through the adjuster and snagging on the seat but will permit easy removal and replacement of the attacment strap assemblies. The strap ends shall be seared to preven fraying.

3.8.9 Dropline assembly. The dropline assembly shall consist of the following: a dropline that will retain the inner container (packed with 35 pounds of survival equipment) and the inflated LRU-16/P raft to the individual during parachute descent and a cable-valve release that will actuate the LRU-16/P raft inflating assembly by the deployment of the dropline cause by the weight of the survival kit container.

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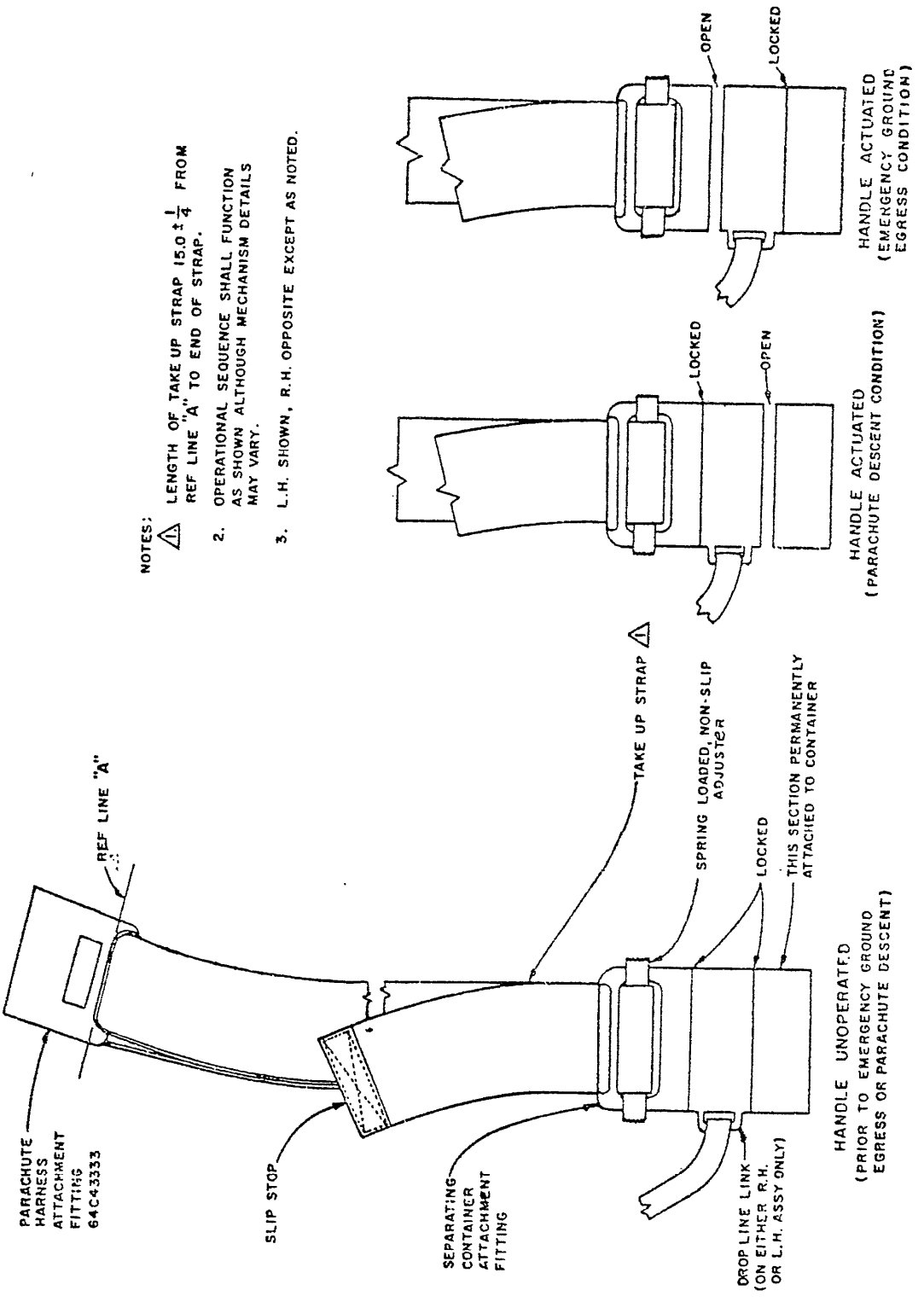


FIGURE 2. Container-to-user hardware.

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3.8.9.1 Dropline. The dropline shall be constructed of webbing in accordance with USS Chemicals Div. Drawing RP 755-32. The webbing shall be 3/4-inch-wide natural-color webbing that conforms to MIL-W-5625. The dropline shall be 25 feet long. The dropline shall have a stop that will prevent deployment of the dropline until the lid is opened and that will cause separation of the lid from the compartment during deployment. The dropline shall have a loop for attaching the raft, The loop, on the flat, shall be 2-3/8 inches (plus or minus 1/8 inch). The loop shall be located 10 feet from the inner container. One end of the dropline shall be permanently attached to the survival kit container separating hardware. The dropline shall be routed under the lid, through the lid and the compartment mating extrusions. The distal end of the dropline shall be attached to the survival kit container, The free length of the dropline shall be reefed, as flat as possible, into the dropline retainers.

3.8.9.2 Cable-valve release. The cable-valve release shall conform to the -5 configuration on Drawing 50B3502. The cable-valve release shall be attached to the dropline so that the ball will be 6-1/4 inches (plus or minus 1/4 inch), measured on flat, from the end of the raft-attaching loop.

3.8.10 Dropline retainers. The retainers for the dropline shall be made of rubberized fabric, The retainers for the dropline shall be attached to the inside of the lid by means of four 2-inch-squares of fastener tape. The hook fastener tape shall be Green 3421 and shall conform to type II, class 2 of MIL-F-21840. The pile fastener tape shall be Green 3421 and shall conform to class 2 of MIL-F-21840.

3.8.11 Dropline connecting plunger device. The dropline connecting plunger device shall be an integral part of the side release hardware so that no cable linkage will be required. Alternate configurations for the dropline connecting plunger device, which meet the functional and performance requirements specified herein, will be considered for approval by the procuring activity.

3.8.12 Override selector device. The override selector device shall permit inflight user selection between automatic and manual operation of the survival kit container after ejection or bailout and shall act as a safety device during ground handling. The override selector device shall have a selector lever. The selector lever shall be on the front of the survival kit container and shall be on or near the right corner in a position that is easily accessible to the aircrewman and that is compatible with the aircraft crew station and the ejection seat. The selector lever shall operate in a vertical plane with the upward position permitting automatic actuation and the downward position permitting only manual operation of the survival kit container. The position of the selector lever shall be recognizable by feel by the aircrewman (wearing flight gloves) The selector lever shall be detented in each position to prevent the selector lever from being moved accidentally or from being moved by acceleration or windblast forces. The positions of the selector lever shall be permanently and adequately marked to describe the positions for both the aircrewman and the maintenance personnel.

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3.8.13 Retention straps. Two retention straps to retain the inner container after deployment and during parachute descent shall be installed diagonally in the survival kit container. The retention straps shall conform to MIL-S-83862.

3.8.14 Deployment sensing cable assembly. The deployment sensing cable assembly shall consist of a cable assembly properly mounted and stowed and with disconnects for compatibility with escape system operation and maintenance. Any interconnections, installation methods, or actuation processes between the survival kit container, the parachute recovery subsystem, the personnel restraint subsystem, or the seat assembly shall not compromise the reliability of seat-man separation or deployment and inflation of the parachute canopy, Accidental or premature actuator operation, while the deployment sensing cable assembly is connected into the seat system, shall not be possible. Any maintenance connection point(s) of the deployment sensing cable assembly shall be easily accessible and visible for preflight inspection and shall be operable without special tools and by personnel wearing heavy gloves.

3.19 Safety safety program. The manufacturer of the survival kit containers shall establish and maintain an effective system safety program that will comply with the applicable requirements of MIL-STD-882.

3.10 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer’s part number and changes thereto.

3.11 Weight. The weight of the survival kit container, including all of the components listed on figure 1, shall not exceed 23 pounds.

3.12 Color. The color of the basic container shall be sage green conforming to 34092 f FED-STD-595. If the basic container is made of molded fiber glass, the glass fibers shall not be considered a PART of the color

3.13 Marking

3.13.1 Decals. Decals to show operation and maintance procedures for the hardware shall be used where necessary.

3.13.2 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance-with MIL-STD-130.

3.14 Workmanship. The survival kit containers shall be made in accordance with the best shop practices for this type of equipment. The survival kit containers shall be free of defects such as burrs, sharp edges, rough grinding marks, and uneven surfaces due to forming operations. All exterior and interior surfaces of the survival kit containers shall be smooth, consistent with other requirements of this specification, to avoid sharp protrusions that might tear, cut, or puncture the contents of the survival kit container during and after its first operation. Fiber glass reinforced plastic materials shall be free of excessive resin buildup, inconsistency of glass fibers, wrinkles, blisters, cracks, starved areas, air bubbles, tackiness, gaps, foreign objects, and laps.

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Particular attention shall be given to accuracy of dimensions; radii of fillets; marking of parts and assemblies; thoroughness of welding, brazing, painting, riveting, swaging, and soldering; alignment of parts; and tightness of assembly screws and bolts. Castings shall be high quality; clean; sound; and free of holes, porosity, cracks, and any other defects,

3.14.1 Alignment. The alignment of fitting and mating surfaces shall be accurate to a degree that will permit the proper functioning of the survival kit container within the environmental conditions expected during the service life of the survival kit container.

3.14.2 Assembly screws and bolts. Assembly screws and bolts shall be tight, The word tight shall mean that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, nut, or components that are being secured by the screw or bolt.

3.14.3 Cleaning. The survival kit container shall be thoroughly cleaned. Any oil and foreign matter that might adversely affect the operation of the survival kit container or the survival equipment with which it comes in contact shall be removed during and after final assembly.

3.14.4 Protection. The survival kit container shall be protected against the entry of all foreign materials.

3.14.5 Lubrication. Permanent-type dry lubrication that will last the expected life of the survival kit container and that meets with the approval of the procuring activity shall be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article tests

(b) Quality conformance tests.

4.3 Test conditions

4.3.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77 (plus or minus 18) degrees Fahrenheit, and at a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from these values, proper allowance shall be made for the change in instrument reading.

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4.3.2. Test attitude. Unless otherwise specified, the survival kit container shall be tested in the normal operating attitude.

4.4 First article tests. The first article tests shall consist of all of the tests described under 4.6 and shall be conducted in accordance with the approved first article test program [see 6.2).

4.4.1 Sequence of tests. Before any other first article tests are performed, the three automatic actuators shall be tested to demonstrate compliance with MIL-C-83124. The first article tests specified in 4.4 shall be conducted in the following sequence:

(a) Examination of product

(b) Environmental tests

(c) Other tests.

4.4.2 First article samples and report. The contractor shall subject ten survival kit containers to the first article tests specified in 4.4 and shall test three automatic actuators as specified in 4.4.1. Two survival kit containers shall be used for all the first article tests except for the reliability tests described under 4.6.14 and the parachute drop test described in 4.6.9. Five survival kit containers shall be subjected to the' reliability testing. Two first article samples, tested or untested, and a first article report prepared in accordance with MIL-STD-831 shall be furnished to the procuring activity (see 6.2).

4.4.3 Sled tests. If approved by the procuring activity, sled tests in accordance with MIL-STD-846 may be performed in lieu of the static tests desribed herein. If the option to use sled tests is requested, the request shall outline the reasons for the request.

4.4.4 Reliability testing. Four first article samples shall be subjected to the reliability tests described in 4.6.14.1. One first article sample shall be subjected to the reliability tests described in 4.6.14.2. Failure of a survival kit container to perform any function during the reliability testing shall constitute one failure. If failures occur during reliability testing, acceptance shall be judged as specified in table II. After one failure, direction-to either continue or repeat the test may be given at the option of the procuring activity.

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TABLE II. Reliability accept-reject criteia

Minimum Test Cycles for Acceptance	Allowable Failures
240	0
390	1
530	2
670	3

4.4.4.1 Longevity testing. Two of the first article samples that have been subjected to the reliability tests described in 4.6. 14.1 shall be continued through the longevity test described in 4.6.14.1.2. Acceptance for the longevity test shall be judged on the basis of 4.4.4.

4.5 Quality conformance tests. Quality conformance tests shall consist of the following:

(a) Individual tests

(b) Sampling tests.

4.5.1 Individual test. Each survival kit container shall be subjected to the following tests as described under 4.6:

(a) Examination of product

(b) Hardware operation.

4.5.2 Sampling tests. Samples shall be s-elected at random from each lot or fraction of a lot and shall be subjected to the sampling tests.

4.5.2.1 Lot. A lot shall consist of 100 units of product that have been manufactured under essentially the same conditions and at essentially the same time. The unit of product for the expression of the lot size for plans A and B shall be one survival kit container. The unit product for the expression of the lot size for plan C shall be one fiber glass basic container before any assembly of hardware. The unit of product for the expression of the lot size for plan D shall be one shipping container full prepared for delivery except for closure of the shipping container.

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4.5.2.2 Plan A. Four survival kit containers from each lot plus 4 percent of any fraction of a lot rounded to the next highest number shall be subjected to the following tests as described under 4,6:

(a) Interchangeability of critical parts

(b) Proof loads (see 4.6.5).

4.5.2.2.1 Reconditioning. After the completion of the-sampling tests specified for plan A, the survival kit containers shall be reconditioned, as required, for delivery as serviceable survival kit containers.

4.5.2.3 Plan B. One survival kit container from each lot and any fraction of a lot (up to a maximum of four survival kit containers per contract or purchase order) shall be subjected to the lid retention test described in 4,6.6.

4.5.2.4 Plan C. If the basic container is made of fiber glass material, one basic container before the assembly of any hardware from each lot and any fraction of a lot (up to a maximum of four basic containers per contract or purchase order) shall be subjected to the fiber glass container test described in 4.6.7.

4.5.2.5 Plan D. Four survival kit containers fully prepared for delivery except for closure from each lot plus 4 percent of any fraction of a lot rounded to the next highest number shall be subjected to the examination of preparation for delivery test described in 4.6.15.

4.5.2.6 Rejection and retest. If one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure have been determined. The contractor shall explain fully to the Government representative and notify the procuring activity in writing the cause of failure, the action taken to preclude recurrence, and the impact this failure may have in scheduled deliveries. After correction, all of the sampling tests shall be repeated.

4.5.2.7 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. Final acceptance of the entire lot or lots produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

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4.6 Test methods

4.6.1 Examination of product. The survival kit container shall be examined visually to determine compliance with the requirements of this specification with respect to materials; finish, weight, identification marking, and workmanship.

4.6.2 Environmental tests

4.6.2.1 High temperatre operation. The survival kit container shall be subjected to a high temperature test in accordance with method 501, procedure I of MIL-STD-810. While the survival kit container is at the high temperature, the hardware of the survival kit container shall be subjected to and pass one test each in accordance with 4.6.3.1 and 4.6.3.2.

4.6.2.2 Temperature shock. The survival kit container shall be subjected to a temperature shock test in accordance with method 503, procedure I of MIL-STD-810. At the end of the specified period, the survival kit container shall be returned to room temperature, operated as specified in 4.6.3.1 and 4.6.3.2 for one test, arid inspected for signs of deterioration and damage.

4.6.2.3 Low temperature operation. The survival kit container shall be subjected to a low temperature test at minus 65 degrees Fahrenheit in accordance with method 502, procedure 1 of MIL-STD-8l0, While the survival kit container is at the low temperature, the hardware of the survival kit container shall be subjected to and pass one test each in accordance with 4.6.3.1 and 4.6.3.2.

4.6.2.4 Dust. The complete survival kit container with carbon dioxide cylinder, raft, and simulated survival equipment shall be subjected to a dust test in accordance with method 510, procedure I of MIL-STD-810. At the end of the first 6-hour test period, step 1, the orientation of the survival kit container shall be changed to another vulnerable position relative to the dust stream. After the dust test, the survival kit container shall be subjected to and pass the tests specified in 4.6.3.1 and 4.6.3,2 for ten alternate operations.

4.6.2.5 Humidity. The survival kit container shall be subjected to a humidity test in accordance with method 507, procedure I of MIL-STD-810 for not less than 120 hours and then shall be subjected to and pass one test each in accordance with 4.6.3.1 and 4.6.3.2.

4.6.2.6 Salt atmosphere. The survival kit container shall be subjected to a salt fog test in accordance with method 509, procedure I of MIL-STD-810. After the salt fog test, the survival kit container shall be subjected to and pass the tests specified in 4.6.3.1 and 4.6.3.2 for ten alternate operations.

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4.6.2.7 Vibration. The complete survival kit container shall be mounted on a vibration stand for 6 hours on each of the three axis, two horizontal and one vertical, operating at a frequency ranging from 5 to 500 Hertz with a double amplitude not to exceed 0.036 inch and a vibratory acceleration not to exceed plus or minus 10 G. After the vibration tests, the survival kit container shall be subjected to and pass one test each as specified in 4.6.3.1 and 4.6.3.2. The survival kit container shall be strapped tightly to the vibration table with l-3/4-inch-wide cotton or polyester-cotton webbing.

4.6.2.8 Shock. The survival kit container shall be subjected to the shock test specified in method 516, figure 516.2-1, procedures I and 111, amplification a and duration c of MIL-STD-810. After the specified testing, the survival kit container shall be subjected to and pass one test each as specified in 4.6.3.1 and 4.6.3.2.

4.6.3 Hardware operation. The operation of the hardware of the survival kit container under the following simulated conditions shall be tested. Failure of the hardware to accomplish any specified function shall be cause for rejection of the survival kit container.

4.6.3.1 Parachute descent. The override selector lever shall be placed in the down Position. The actuator device shall be operated simulating parachute deployment and seat separation. The deployment sensing cable assembly shall operate and separate from the survival kit container by a force not to exceed 10 pounds applied at the distal end of the seat attachment lanyard in random directions to simulate seat-man separation (type I survival kit container) or a force not to exceed 50 pounds applied at the canopy riser fitting in random directions to simulate canopy riser deployment from the parachute pack (type II survival kit container). The automatic actuator shall not be initiated. The dropline connecting plunger device shall be extended simulating seat separation. The operating handle of the survival kit container shall unlock with a pull of 20 pounds (plus or minus 5 pounds). Separation of the survival kit container hardware, attachment of the dropline, and opening of the lid lock shall then occur when a maximum pull of 15 pounds, except when subjected to the tests described under 4.6.2, is applied to the operating handle. The maximum pull for the tests described under 4.6.2 shall be 30 pounds. The operating handle shall separate completely from the survival kit container when the maximum force is applied. A 50-pound internal upward force shall be applied to the center of the survival kit container lid. A 25-pound force at 120 degrees to the top of the compartment shall be applied at both sides of the survival kit container at the attaching hardware. This test shall be repeated for a total of five tests.

4.6.3.2 Emergency ground egress. The dropline connecting plunger device shall be depressed to simulate installation in an ejection seat. The operating handle of the survival kit container shall unlock with a pull of 20 pounds (plus or minus 5 pounds). Separation of the survival kit container shall then occur

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when a maximum force of 15 pounds, except during and after the tests described under 4.6.2, is applied to the operating handle. During and after the tests described under 4.6.2, the maximum force shall be 30 pounds, The operating handle shall separate completely from the survival kit container when this force is applied. The dropline shall be locked to the portion of the hardware remaining with the survival kit container, This test shall be repeatedfor a total of ten operations, A 15-pound load applied in accordance with 4,6,13,1,2 shall be used on each parachute attachment, and a 50-pound internal upward fore shall be applied to the center of the survival kit container lid.

4.6.3.3 Automatic operation, The override selector lever shall be moved to the up position. A 50-pound upward internal force shall be applied to the center of the lid. An upward 25-pound force, directed 30 degrees aft of normal to the top of the lid, shall be applied to each of the survival kit container attachment straps. The dropline connecting plunger device shall be extended, simulating seat separation. The deployment sensing cable assembly shall be operated in a random direction simulating conditions during seat system ejection The deployment sensing cable assembly shall operate and separate from the automatic actuator by a force not to exceed 10 pounds applied at the distal end of the seat attachment lanyard in random directions to simulate seat-man separation (type I survival kit container) or a force not to exceed 50 pounds applied at the canopy riser fitting in random directions to simulate deployment of the canopy risers from the parachute pack (type 11 survival kit container) . The automatic actuator shall fire the cartridge, and the survival kit container hardware shall be operated 4 seconds (plus or minus 1 second) later. The operating mechanism shall be detented in the operated position. The cartridge may be simulated for the quality conformance tests only.

4.6.4 Interchangeability of critical parts. Functional interchangeability of the lids, the cushions, the operating handles, and the survival kit container attaching hardware shall be determined by interchanging each item on four survival kit containers and actuating for one test each in accordance with 4,6.3.1 and 4.6.3.2.

4.6.5 Proof loads, The proof loads of 4.6.13.3.1, 4.6,13,2, and 4,6,13.4 shall be applied to the survival kit container. The survival kit container shall then be subjected to and pass one test each as specified in 4.6.3.1 and 4.6.3.2 and shall be visually inspected. There shall not be any damage or permanent deformation to the survival kit container.

4.6.6 Lid retention. The lid shall be subjected to an upward load of 700 pounds normal to the lid at the center. The load shall be applied to a cable or webbing threaded through a hole or slot in the lid and attached to a 6-inch-diameter metal plate that bears against the underside of the lid. The test shall be repeated with the load applied 3 inches to the right of the center point and then 3 inches to the left of the center point. There shall not be

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any evidence of failure of the lid locks or damage or permanent deformation. The survival kit container shall then be subjected to and shall pass one test each as specified in 4.6.3.1 and 4.6.3.2.

4.6.7 Fiber glass container. Four coupons of the fiber glass material shall be taken randomly from the basic container and tested in accordance with the applicable methods of FED-STD-406 to determine compliance with the requirements specified in 3.4.2.1.

4.6.8 Parachute opening shock. The survival kit container shall be subjected to simulated parachute opening shock by means of a drop test. The drop test shall consist of dropping the loaded survival kit container to the end of a suspension assembly of such height and elasticity as to give the required shock. The survival kit container shall be loaded with ballast, simulating survival equipment, to achieve a total weight of 50 pounds. The loaded survival kit container shall be secured at the parachute attaching hardware on both sides to a webbing assembly (simulating the parachute harness) which, in turn, is attached to a drop cable. The length and elasticity of the drop system shall be such as to achieve a force of at least 30 G (peak) and a minimum rate of increase of 500 G per second between the webbing assembly and the drop cable. Following this test, the survival kit container shall be operated to determine compliance with 4.6.3.1 and 4.6.3.2.

4.6.9 Parachute drop. Ten parachute drop tests shall be conducted at various speeds using articulated dummies, parachutes, ejection seat (if appropriate) and survival kit containers that are packed to simulate operational use. The speeds and altitudes of the drops shall vary from zero knot at 2,000 feet altitude through 400 knots at 18,000 feet altitude (parachute deployment 1 second after aircraft exit). The override selector lever shall be in the upward position for automatic operation of the survival kit container. The actuator shall operate the survival kit container hardware. The type I survival kit container shall deploy on its dropline 4 seconds (plus or minus 1 second) after seat-man separation. The type II survival kit container shall deploy on its dropline 4 seconds (plus or minus 1 second) after the canopy risers deploy from the parachute pack. During these tests there shall not be any damage to or decreased performance of the parachute, the seat, or the survival kit container or damage to the dummy which can be attributed to the actuator. Failure to meet the preceding conditions as specified shall be cause for rejection.

4.6.10 Windblast. The survival kit container shall be placed in a simulated ejection seat bucket and shall be attached to a personnel restraint harness being worn by a 95th percentile anthropomorphic dummy. The survival kit container shall be subjected to a Windblast of 600 knots at sea level or equivalent. The wind tunnel shall be equipped with a deflector and shutoff so that the rate of onset of the windblast is not more than 0.12 second. Shutoff of the wind blast shall start immediately after the deflector is in an open position. Decay shall

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Occur in not less than 1 second, The lid shall not separate from the survival kit container. The operating handle shall remain on the survival kit container, There shall not be any visible deformation to the survival kit container, Upon completion of the windblast test, the survival kit container shall be subjected to and pass one test each as specified in 4.6.3.1 and 4.6.3.2,

4.6.10.1 Ejection seat survival kit container separation. Windblast tests may be conducted in conjunction with ejection seat survival kit container separation tests at the maximum speed specified in MIL-STD-846. The survival kit container its attaching hardware and components shall withstand this windblast without any damage or structural failure.

4.6.11 Magnetic interference. The survival kit container shall be placed 6 inches from the face of a pilot's standby compass that conforms to MS17983 or equivalent. The survival kit container shall not deflect the compass more than 3 degrees.

4.6.12 Adjuster

4.6.12.1 Adjuster creep. A cycle consisting of alternating loads of 20 pounds and zero pound shall be applied for 30 circles in each of the following five directions: normal, 60 degrees forward, 60 degrees aft, and 60 degrees to either side of normal to the survival kit container lid. The test shall be started by pulling the webbing through the adjuster to eliminate any set in the webbing or false load of the spring-loaded slider. The 20-pound load and the zero-pound load shall be at a stroke of at least 3/4 inch. At the end of the 90 cycles, the webbing shall not have slipped more than 1/2 inch,

4.6.12.2 Release of adjuster. An initial load of 20 pounds of tension shall be applied in the attachment webbing. A load of 30 pounds shall be applied to the slider assembly. When the 30-pound load is applied to the slider-assembly, the adjuster shall allow complete slippage of the webbing.

4.6.13 Static loads

4.6.13.1 Parachute attaching hardware

4.6.13.1.1 Hardware and container section. The parachute attaching hardware and the survival kit container section to which it is attached shall be subjected to load conditions of 1,400 pounds (proof) and 2,100 pounds (ultimate) for each side. The proof loads shall be applied in segments of 45 degrees in a 30-degree done about each attachment. The ultimate loads shall be applied to each attachment in the outboard segment only.

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4.6.13.1.2 Handle operation. A load of 100 pounds shall be applied to the parachute attaching hardware at a 30-degree angle towards the front, then aft, and to each side of the survival kit container hardware. Load shall be applied 2 inches from the forward end of the operating handle and shall move through the arc of the handle travel. A pull of less than 40 pounds shall operate the handle and release the survival kit container hardware both under emergency ground egress and parachute descent operating conditions.

4.6.13.2 Container (seating section). The basic container shall be placed with the top surface of the container horizontal to the compression tester so that the load will be applied vertically. A compression test form, consisting of a hard rubber block and having an outline as described on figure 3, shall be placed on the container and oriented as shown. Compression loads of 4,500 pounds proof and 5,600 pounds ultimate shall be applied to the compression test form for a period of 7 seconds. The loads shall be applied vertically and upward compressing the container between the base of the test stand and the force apparatus. The container shall not show any structural failure or permanent deformation after the 4,500-pound load is relieved. The ultimate strength of the container at the section mentioned above shall be 5,600 pounds. Distortion or warpage will be permitted after the 5,600-pound ultimate load is relieved. After this test, the body of the container, the lid, the closures and the hardware fittings shall function properly in accordance with 4.6.3.1 and 4.6.3.2.

4.6.13.3 Handle side load. The handle shall be installed in the normal locked position. An outward proof load of 500 pounds shall be applied to the top of the rigid portion at the front of the handle parallel to the surface of the lid for a minimum period of 10 seconds. The handle shall not show any permanent deformation or damage.

4.6.13.4 Handle lock load. A proof load of 200-pound tension shall be applied to the grip of the handle perpendicular to the surface of the lid for a minimum period of 10 seconds. The handle and the lock shall not show any permanent deformation or damage.

4.6.13.5 Dropline connecting plunger device. A load of 200 pounds shall be applied to the lowest point of the dropline connecting plunger device in the extended position in both fore and aft directions and from both sides of the survival kit container. After the application of the 200-pound load in each of the specified positions, the survival kit container shall be subjected to and pass one test each as specified in 4.6.3.1 and 4.6.3.2. The actuation of the dropline connecting connecting plunger device shall be determined and shall be as specified in 3.7.12.5.

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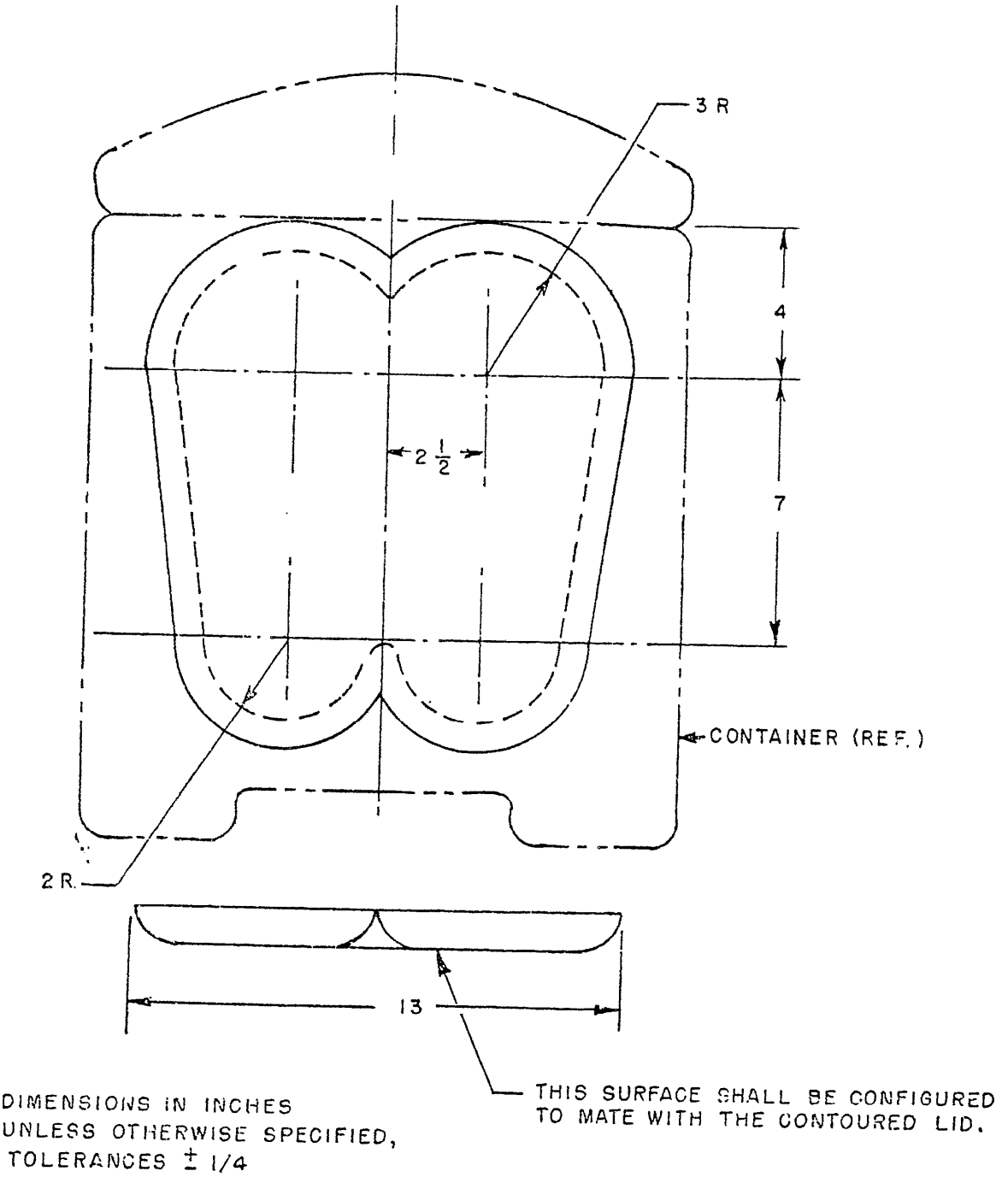


FIGURE 3. Compression test form

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4.6.14 Reliability Five survival kit containers that have been designated units 1 through 5 shall be subjected to the reliability tests as specified herein to demonstrate the reliability specified in 3.7.13. Except for the new sensing cables specified in 4.6.14.1.1, no parts of the survival kit containers shall be replaced as preventive maintenance during the reliability tests.

4.6.14.1 Manual operating mode. The four survival kit containers that have been designated units 1 through 4 shall be subjected to tests as specified in table III. The hardware of-the survival kit--containers shall be cycled in accordance with the schedule specified in table III for a minimum of 240 consecutive successful cycles. The survival kit containers shall be loaded and operated as specified by 4.6.3.1. Afailure shall result when one of the functions specified, in 3.7.13.1 does not occur.

TABLE III. Distribution of reliability test cycles

Test Methods	Cycles Per Test				
	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
Dust (see 4.6.2.4)			1/	10	2/
Humidity (see 4.6.2.5)			1/	10	2/
Salt atmosphere (see 4.6.2.6)			1/	20	2/
Vibration (see 4.6.2.7)			1/	20	2/
Parachute descent (see 4.6.3.1)	60 3/	60	60		
Automatic operation (see 4.6.3.3)					10

1/ The unit shall be subjected to the environmental test without operational cycling before exposure to the next test.
2/ The unit shall be subjected to the environmental test and then subjected to one automatic operation as specified in 4.6.3.3. The unit shall be reconditioned after the automatic operation before it is subjected to the next test.
3/ Unit 1 shall be subjected to ten of these cycles as specified in 4.6.14.1.1.

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4.6.14.1.1 Sensing system. Ten of the sixty cycles specified in table III for the survival kit container designated unit 1 shall be conducted by dropping a 2-pound weight 25 feet while it is attached to a steel cable connected to the sensing system. The sensing cables shall be extracted without breakage. The sensing system shall operate as designed without damage. Anew sensing cable shall be used for each of the ten cycles.

4.6.14.1.2 Longevity. The survival kit containers that have been designated units 1 and 2 shall be continued through a longevity test after completion of the reliability tests. One of the survival kit containers shall be-operated as specified in 4.6.3.1 for 240 cycles. The other survival kit container shall be operated as specified in 4.6.3.2 for 240 cycles. No cable adjustments shall be made or required during the longevity test. Parts requiring replacement within this period of time and the normal operating period of such parts shall be recorded by the contractor.

4.6.14.2 Automatic operating mode. The reliability of the automatic actuato shall be in accordance with MIL-C-83124 and shall be demonstrated during the static firing tests specified therein. In addition, the survival kit container that has been designated unit 5 shall be subjected to tests as specified in table III. The hardware of the survival kit container shall be cycled in accordance with the schedule specified in table III for 14 consecutive successful cycles, The survival kit container shall be loaded and operated as required in 4.6.3.3. A failure shall result when one of the functions specified in 3.7.13.2 does not occur.

4.6.15 Examination of preparation for delivery. The shipping container fully prepared for delivery with the exception that it need not be sealed shall be examined to determine conformance to good workmanship and to the levels of preservation and packaging and of packing specified in the contract or order.

4.6.16 Maintainability verification. Maintainability verification shall be in accordance with test method 3 of MIL-STD-471.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be in accordance with MIL-STD-794 and shall be level A or C as specified (see 6,2).

5.1.1 Level A. Each survival kit container shall be preserved in accordance with method III of MIL-P-116.

5.1.2 Level C. Each survival kit container shall be packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. Level C may conform to the supplier's commercial practice provided the latter meets the requirements for this level.

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5.2 Packing. Packing shall be level A, B, C or as specified (see 6.2).

5.2.1 Level A. Survival kit containers preserved and packaged as specified shall be packed in accordance with level A of MIL-STD-794. The gross weight of each pack shall be limited to approximately 200 pounds,

5.2.2 Level B. Survival kit containers preserved and packaged as specified shall be packed in accordance with level B of MIL-STD-794. The gross weight of each pack shall be limited to approximately 500 pounds,

5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be packed in accordance with level C of MIL-STD-794.

5.3 Marking In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with the require. ments of MIL-STD-129.

6. NOTES

6.1 Intended use. The survival kit containers covered by this specification are intended to be used in aircraft that are equipped with upward ejection seats.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Type required (see 1.2)

(c) First article requirements, including first article test program and first article test report (see 3.1 and 4.4 through 4.4.4.1)

(d) Any major component not required (see 3,2)

(e) Aircraft for which ejection seat is intended (see 3.5)

(f) Survival equipment requirements (see 3,5.2)

(g) List of interchangeable components and assemblies (see 3,7.3)

(h) Requirements for maintainability report (see 3.7.14)

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(i) If lid is to be flat or contoured and if lid is not to be retained to dropline during survival kit container deployment (see 3.8.1.2]

(j) Selection of applicable levels of preservation and packaging and packing (see 5.1 and 5.2).

6.3 Color samples. Color samples can be obtained from the procuring activity or as directed by the contracting officer.

Custodian:
Air Force - 11

Reviewer:
Air Force - 99

Preparing activity:
Air Force - 11

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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